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Primary electron drift in a volume hybrid multicusp H^- ion source

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A new, large, hybrid volume negative ion source is described. Observations of curvature and *grad B* primary electron drift in the multicusp magnetic field are reported. The direction of the electron drift changes from one filament to the neighboring one, because of the change in the direction of the magnetic field. For a given discharge current the extracted negative ion current and the plasma density at the center of the extraction region are affected by the direction of the primary electron drift of the active filaments. It was shown, however, that the negative ion current is controlled by the plasma density at the center of the extraction region. A trajectory calculation, effected in the guiding center approximation, allows to estimate the average drift velocity.

I. INTRODUCTION

Primary electron drift related to the curvature and gradient of the multicusp magnetic field has been observed in the hybrid negative ion source. The existence of this drift, parallel to the cusp lines, interferes with the mirror trap effect and has to be taken into account in the estimate of the primary electron lifetime.

In the hybrid multicusp source the filaments are located in the multicusp magnetic field in the median plane between two magnets. It was observed that, in agreement with theory, the direction of the drift changes from one filament to the neighboring one, because of the change in the magnetic field direction. It is shown that, depending on the drift direction of the produced primary electrons, some filaments are more efficient in plasma production than others. Based on these observations, it might be reasonable to reduce the number of filaments in this source maintaining only the more efficient ones. The reduction of the number of filaments is beneficial to negative ion beam production.[1,2]

These observations can be relevant to other types of multicusp ion sources, since it is known that, even when produced by filaments located in the magnetic field-free region, primary electrons end by being trapped in the multicusp magnetic field.[3]

The effort in studying volume negative hydrogen ion sources at Ecole Polytechnique has been continued by building in 1990 a scaled up version of the hybrid volume source.[4] Several purposes have been pursued: (1) study a large source, as requested for fusion applications; (2) generate negative ion beams at lower gas pressure than it is possible in small sources; (3) install a sufficient number of large openings to accommodate windows and probes, used in plasma diagnostics.

II. EXPERIMENTAL SETUP

The new hybrid source is a 40-ℓ cylindrical stainless-steel chamber with an inner diameter of 44 cm and a height of 35.5 cm (Fig. 1). Sixteen Sm–Co permanent magnets are mounted in tubes, welded to the inner side of the cylindrical chamber. The magnets are cooled by the water flowing through these tubes. Due to their orientation (alternatively north and south towards the plasma) the magnets produce a multicusp magnetic field. The value of the magnetic field at the surface of the magnets is 3500 G; the radial variation of the tangential component of the magnetic field in the median plane between two neighboring magnets is shown in Fig. 2.

Sixteen filaments are mounted on the upper flange of the source, on a diameter of 33 cm; each filament is located in the median plane between two neighboring magnets. The electrons emitted by the hot-tungsten filament are accelerated to eV_d (V_d is the discharge voltage applied between the filaments and the chamber walls, see Fig. 1) in the sheath surrounding the filaments. The primary electrons are thus confined in the multicusp magnetic field. The source has been operated dc with a power up to 6 kW. Each of the 16 tungsten filaments (0.5 mm in diameter) is heated independently by a separate winding of a transformer. The discharge power is varied by modifying the filament heating current. The large hybrid source can be operated at much lower pressure compared to the previous, smaller, one.[4,5] The range covered in the present experiments is 1–8 mTorr. The optimum pressure range is 2–3 mTorr.

The possibility to observe various plasma regions through the large windows helped us to confirm and study an earlier observation, relative to the primary electron drift occurring in the multicusp magnetic field, where the filaments are located. The intriguing fact is that the direction of this drift changes from one filament to the neighboring one.

The hybrid source[4] contains three distinct regions: (i) a driver region, located near the walls, in which the filaments are located; (ii) an extraction region, which extends over all the central, field-free region; (iii) a region with high n_-/n_e near the extraction aperture. This latter part of the extraction region is magnetized and due to this and to a small positive bias of the plasma electrode, large densities of volume-produced negative ions are concentrated in this region.[5] The ions are extracted through an 8-mm-diam

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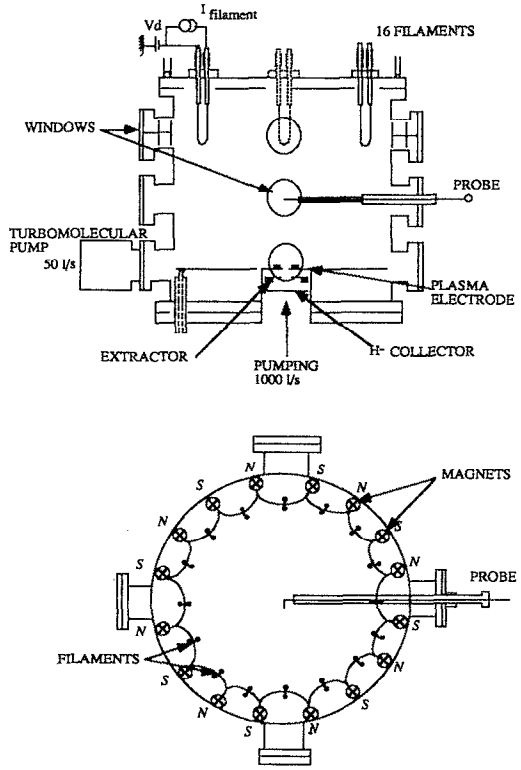


FIG. 1. Schematic presentation of the hybrid negative ion source. In the axial cross section (upper part) the tubes containing the magnets are not shown.

opening in the plasma electrode. The extracted electrons are deflected to the first electrode of the extractor by a magnetic field due to two permanent magnets located in this electrode. The negative ions are weakly deflected by this magnetic field and measured on the collector (see Fig. 1).

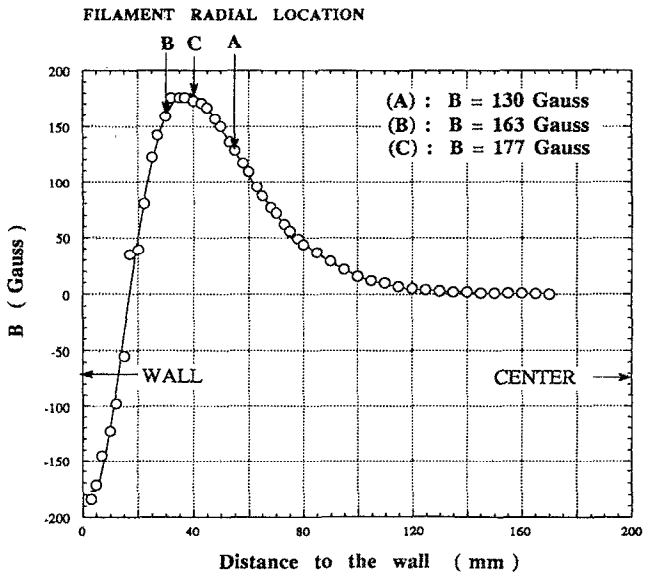


FIG. 2. The variation of the of the tangential component of the magnetic field upon the distance from the wall, in a radial median plane between two neighboring magnets. The filaments are located in this plane and their usual position is indicated by the arrow A.

III. THE DRIFT OF THE PRIMARY ELECTRONS IN THE MULTICUSP MAGNETIC FIELD

As already mentioned the 16 filaments are mounted on a circle (33 cm in diameter). They are located at 5.5 cm from the chamber wall. Their usual position is indicated by the arrow A in Fig. 2, where the dependence of the tangential component of the magnetic field upon the distance from the wall is plotted, in a median radial plane. Note that they are located in a magnetic field of approximately 130 G, and the radial gradient of the magnetic field at this location is 40 G/cm. The electrons emitted by the filaments are injected in the plasma with the energy eV_d, but with a variable radial-velocity component $v_\perp$, perpendicular to the magnetic field. Note that the multicusp magnetic field changes its direction from one filament to the neighboring one. Since the *grad B* has the same sign for all the filaments in position A, the primary electron drift velocity related to the magnetic field curvature and gradient changes its sign when we pass from one filament to the neighboring one. The value of the drift velocity is[6,7]

$$\mathbf{v}_D=\mathbf{v}_{\nabla B}+\mathbf{v}_R=\frac{m}{2q}\left[v_\perp^2+2v_\parallel^2\right]\frac{\mathbf{B}\times\nabla\mathbf{B}}{B^3}. \tag{1}$$

Here $\mathbf{v}_{\nabla B}$ is the *grad B* drift velocity, $\mathbf{v}_R$ is the curvature drift velocity, $\mathbf{v}_\perp$ is the electron velocity perpendicular to the magnetic field, and $\mathbf{v}_\parallel$ is the electron velocity parallel to the magnetic field. With $V_d=100$ V and $v_\perp=v_\parallel=4.2\times10^8$ cm/s, the drift velocity of the 100-eV primary electrons is $v_{D\,\max}=3.7\times10^7$ cm/s.

Thus the filaments are forming two groups: eight filaments (denoted U filaments) producing upstreaming primary electrons, and the eight others (denoted D filaments) producing downstreaming primary electrons. We visually observed light from the gas excitation under the D filaments, while no such light was visible under the U filaments. To quantify the difference between the two types of filaments we moved an electrostatic probe on a circle under the filaments and measured the floating potential and the density of energetic electrons. Figure 3 shows the variation of the floating potential. A large difference is observed between the U filaments (designated by the numbers 1, 3, 5, 7, 9) and D filaments (2, 4, 6, 8, 10). The floating potential varies from -15 to $+2$ V when we move the probe from a D to a U filament, confirming that energetic electrons are found only below the D filaments.

IV. EFFICIENCY OF FILAMENTS WITH DIFFERENT DRIFT DIRECTION

Three filament configurations are compared at constant discharge power: a configuration with all 16 filaments, a second one with only eight U filaments and a third one with eight D filaments. The result is shown in Figs. 4 and 5 where the extracted negative ion current (Fig. 4) and the electron density in the center of the extraction region (Fig. 5) are plotted versus the discharge

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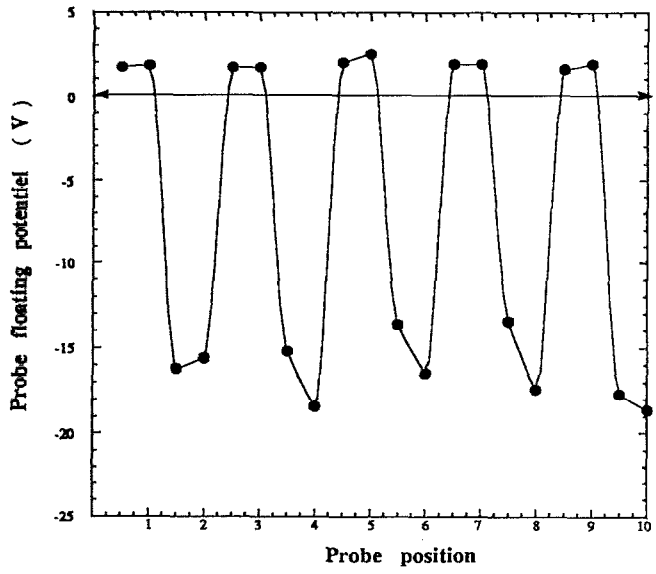


FIG. 3. The floating potential variation when moving the probe on a circle under the filaments, at a distance of 9.5 cm from the filament tips. Pressure: 0.9 mTorr; 50-V, 3-A discharge. The filament position is indicated by the figures (1 to 10) on the "Probe Position" axis.

current I_d, which was produced either from 16 filaments, or from eight filaments. Note that the discharge current generated by each filament is twice as large when eight filaments are used, compared to the case when 16 filaments are used. The eight filaments, not used in a given experiment, are not heated, but remain biased to $-Vd$. It can be noted from Figs. 4 and 5 that the D filaments are 3.5 times more efficient compared to the U filaments.

The electron density at the center of the extraction region (i.e., on the axis at 12.5 cm from the plasma electrode) appears to be determining the extracted negative ion current. In Fig. 6 the extracted negative ion current is plotted as a function of the electron density in the center,

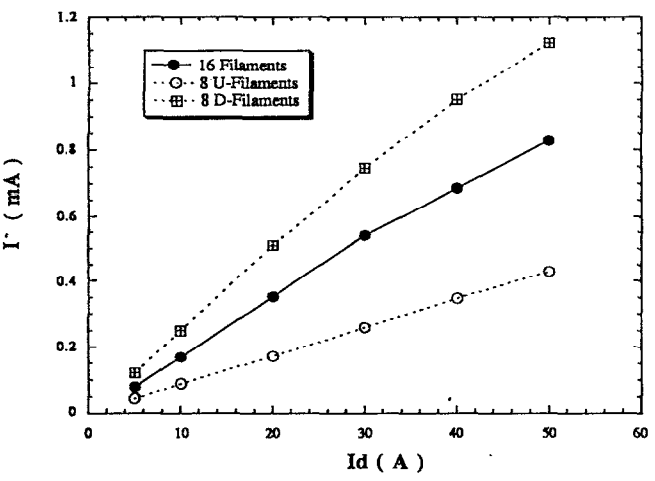


FIG. 4. Variation of the extracted negative ion current with discharge current. Comparison of the cases when 16 filaments, 8 U filaments and 8 D filaments are used. Hydrogen pressure 3 mTorr; discharge voltage: 50 V, extraction voltage 2 kV.

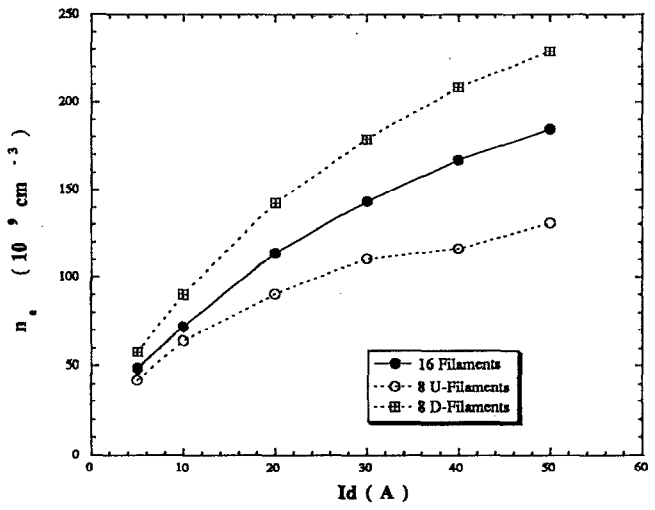


FIG. 5. Variation of the electron density in the center of the source extraction region with the discharge current. Comparison of the cases when 16 filaments, 8 U filaments and 8 D filaments are used. Hydrogen pressure 3 mTorr; the discharge voltage is 50 V.

for the three-filament configurations shown in Figs. 4 and 5. It can be noted that in Fig. 6 the difference in extracted current for the three different configurations is small for the same electron density at the center.

A. Effect of filament radial position

It is obvious that the D filaments are more efficient in plasma production. We assume that this is due to the longer electron lifetime when electrons are drifting down. In the case of electrons drifting up, there are the filament holder leadthroughs which can perturb the electron movement. In order to improve the efficiency of the U filaments, we attempted to change the sign of the electron drift. To achieve this, we studied the effect of the radial position of one of the U filaments. At first a U filament was moved 2.5

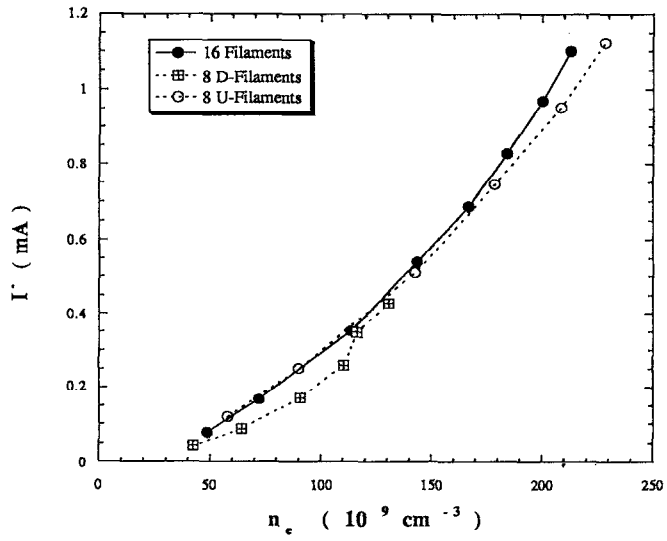


FIG. 6. Variation of the extracted negative ion current with the electron density at the center of the source extraction region. Hydrogen pressure: 3 mTorr; discharge voltage: 50 V.

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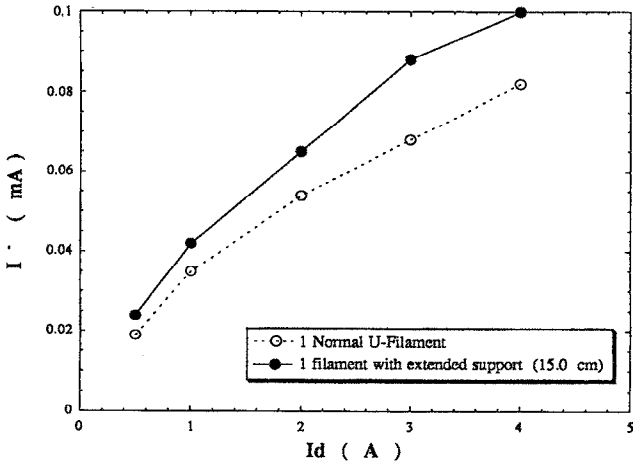


FIG. 7. Variation of the extracted negative ion current with discharge current, with only one filament active. We compare results with a U filament in its original position, with those from a U filament which has been moved by 15 cm toward the extractor by extending its support. Hydrogen pressure: 5 mTorr; discharge voltage: 50 V; extraction voltage: 2 kV.

cm towards the wall (position indicated by arrow B in Fig. 2), reversing in this way the direction of the magnetic field gradient. Measurements were effected with only one filament under operation. A very small improvement was obtained, because with the filament in this position the primary electrons were not drifting down, but moving along the $B=0$ line present nearby.

A third radial position tested was at B_{max}, indicated by the arrow C in Fig. 2. In this position the gradient of the magnetic field is very low; visually we did not observe the light related to the drift of downstreaming electrons. It was not possible to operate the source with only one filament at the low pressure (<2 mTorr) which we have used in the earlier tests. This may indicate that the presence of the primary electron drift helps the low-pressure operation of the source.

B. Effect of filament distance with respect to the extraction aperture

As can be seen in Fig. 1, the filaments were originally installed rather close to the upper flange. The distance of the filament tip to the extraction plane was originally 23 cm (compared to the total distance between the extraction plane and the uper flange, which is 35.5 cm). In a series of experiments we moved one U filament vertically by 15 cm, reducing its distance to the extraction aperture to 8 cm. As a result, a negative ion current increase of 25% was obtained, when the source was operated with only this U filament active (Fig. 7).

We have tested another vertical position of a U filament by moving it by 18.5 cm towards the extraction plane (instead of 15 cm in the first test) with respect to the original one. However, in this position the U filament produced the same extracted negative ion current as in the original position. Thus it appears that there is an optimum distance between the filament and the extraction aperture.

As a result of these tests we moved four U filaments 15 cm closer to the extraction aperture plane. The position of the other four U filaments was not changed, since probe holders were located under these filaments. Our experience is that when a down drifting electron beam crosses a grounded probe holder, it damages its metal screen.

V. MODELING OF ELECTRON TRAJECTORIES

The magnetic field produced by the permanent magnets is calculated by summing the fields created by each of the magnets forming the multicusp magnetic field. The magnetic field of a magnet is calculated considering a uniform magnetization inside the magnet. The agreement with the measurements is excellent. The magnetic field is time independent, but spatially nonuniform in strength as well as in direction.

Most magnetic field lines which exit from the pole of one magnet enter the opposite poles of neighboring magnets. Nevertheless, some lines terminate on the opposite pole of the same magnet. The hot thermionic filaments are located in the median plane between two magnets near the location of maximum field strength (arrow A in Fig. 2).

With a uniform magnetic field, the electron trajectories would be helical along the magnetic field lines. In the nonuniform magnetic field the trajectories are modified by several effects:[7] (1) The mirror effects, related to the magnetic moment conservation due to which the parallel velocity changes along the trajectory. (2) The *grad B* drift, due to the gradient of the magnetic field strength. (3) The curvature drift, due to the spatial variation of the magnetic field direction.

The first effect tends to mirror trap electrons on a magnetic field line. The last two effects create a transverse drift parallel to the cusp lines. All three effects are taken into account in the guiding center approximation, which is moreover their theoretical foundation. In this type of approximation it is possible to calculate the trajectories of primary electrons. This approximation is valid when the magnetic field variation along an electron's gyro-orbit is small permitting a linear development of the theory. Therefore the guiding center approximation is better for lower energy electrons.

The transverse drift velocity is given by Eq. (1), while the evolution of the parallel velocity of the guiding center is given by

$$\frac{dv_{\parallel}}{dt}=\frac{\mu}{m}\frac{\mathbf{B}\cdot\nabla\mathbf{B}}{\mathbf{B}}, \tag{2}$$

where m and μ are, respectively, the electron mass and its magnetic moment, which stays constant along the trajectory. From these two equations and direct calculation of B and *grad B* in each point, the guiding center trajectory is calculated using a predictor-corrector method.

The trajectory calculation confirms that, if the initial magnetic moment is large enough to permit reflection in the magnetic mirror between two contiguous magnets, the primary electrons are confined and oscillate along the magnetic field line between the turning points. The mirror ratio

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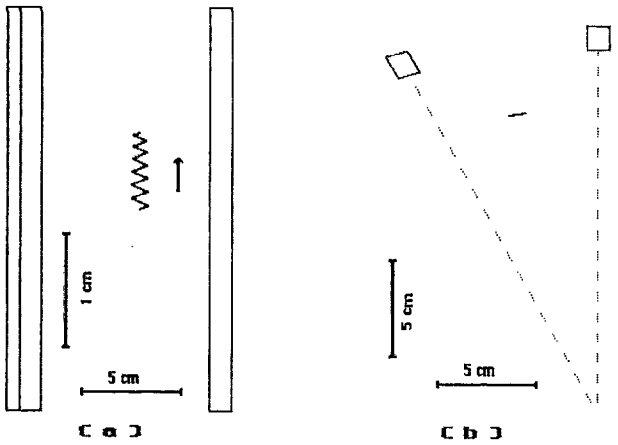


FIG. 8. Guiding center trajectory calculated for a 100-eV electron emitted from the filament position, with initial $v_{\perp 0}=v_{\parallel 0}$. Two projections are shown (a) on a vertical plane, perpendicular to the magnetization of the right-hand magnet; (b) on a horizontal plane.

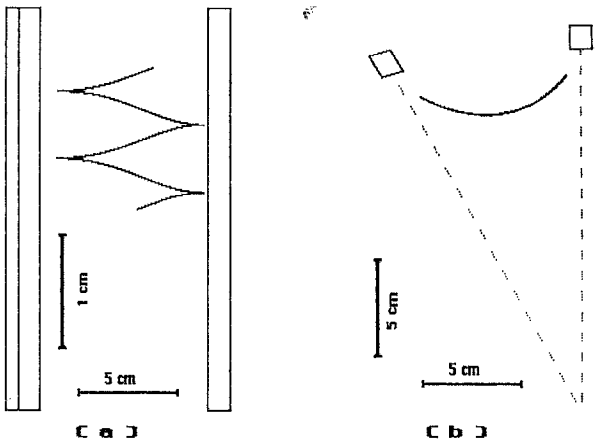


FIG. 9. Guiding center trajectory calculated for a 100-eV electron emitted from the filament position, with initial $v_{\perp 0}/v_{\parallel 0}=10$. Two projections are shown (a) on a vertical plane, perpendicular to the magnetization of the right-hand magnet; (b) on a horizontal plane.

B_{max}/B_{min} is in the range between 22 and 27 ($B_{max}=3500$ G on the magnet and B_{min} depending on the radial position of the filament in the median plane between two magnets). The transverse drift is added to this oscillatory movement. In agreement with observations, the main transverse drift occurs in a direction parallel to the cusp lines, related with the *grad B* and curvature. This drift is inverted in a north–south zone compared to a south–north zone.

Figure 8 presents the guiding center trajectory calculated for a 100-eV electron emitted from the filament position, with initial velocities $v_{\perp 0}=v_{\parallel 0}$. The electron drifts vertically 0.66 cm during 31 ns, which corresponds to an average drift velocity of 2.1×10^7 cm/s. The total path length is 18.4 cm. The excursion along the field lines is very short in this case. With the present ratio $v_{\parallel}/v_{\perp}=1$, the electrons are confined when the mirror ratio is 2.

Figure 9 presents the guiding center trajectory calculated for a 100-eV electron emitted from the filament position, with initial velocities $v_{\perp 0}/v_{\parallel 0}=10$. The electron drifts vertically 1.22 cm during 74 ns, which corresponds to an average drift velocity of 1.65×10^7 cm/s. The total path length is 44 cm. Due to the larger $v_{\parallel}$ the excursion along the field lines is now larger.

The calculation does not include collisional effects. The mean free path of primary electrons is generally larger than the distance between magnets. The collisions would act in two ways: (i) facilitating the loss of electrons to the magnets by partly destroying the mirror trapping effect; (ii) adding a random transverse diffusive velocity to the transverse drift velocity. The self-consistent electric field in the magnetized region can also modify the drifts and should also be taken into account in a more elaborate calculation. The guiding center calculations are in agreement with experimental observations. However, if collisional effects and electric field effects are not included, only qualitative agreement can be expected.

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